

SpikeStereoNet: A Brain-Inspired Framework for Stereo Depth Estimation from Spike Streams

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Abstract

Conventional frame-based cameras often struggle with stereo depth estimation in rapidly changing scenes. In contrast, bio-inspired spike cameras emit asynchronous events at microsecond-level resolution, providing an alternative sensing modality. However, existing methods lack specialized stereo algorithms and benchmarks tailored to the spike data. To address this gap, we propose SpikeStereoNet, a brain-inspired framework and the first to estimate stereo depth directly from raw spike streams. The model fuses raw spike streams from two viewpoints and iteratively refines depth estimation through a recurrent spiking neural network (RSNN) update module. To benchmark our approach, we introduce a large-scale synthetic spike stream dataset and a real-world stereo spike dataset with dense depth annotations. SpikeStereoNet outperforms existing methods on both datasets by leveraging spike streams' ability to capture subtle edges and intensity shifts in challenging regions such as textureless surfaces and extreme lighting conditions. Furthermore, our framework exhibits strong data efficiency, maintaining high accuracy even with substantially reduced training data.

1 Introduction

Depth perception is fundamental for navigating and interacting with the 3D world, driving applications from robotic manipulation [1–4] to autonomous navigation [5–11]. Traditional stereo vision estimates depth from calibrated image pairs captured by frame-based cameras; however, it suffers from motion blur and latency in dynamic scenes. Biological vision systems efficiently process visual inputs using sparse, asynchronous spikes, achieving remarkable speed and energy efficiency [12–18]. Neuromorphic spike cameras [19–27] implement this principle in hardware, delivering ultra-high temporal resolution (up to 40,000 Hz) and capturing rich luminance information through asynchronous binary spike streams. This makes them suitable for robust perception tasks, such as stereo depth estimation, particularly in highly dynamic scenarios where conventional methods are ineffective.

Despite their potential, spike cameras present distinct challenges for stereo depth estimation. Their asynchronous, binary, high-throughput streams conflict with frame-based algorithms that expect synchronous, intensity-valued image pairs [28–42], and any conversions to frames introduce temporal quantization errors, motion blur, or significant computational overhead [23, 25, 43], diminishing

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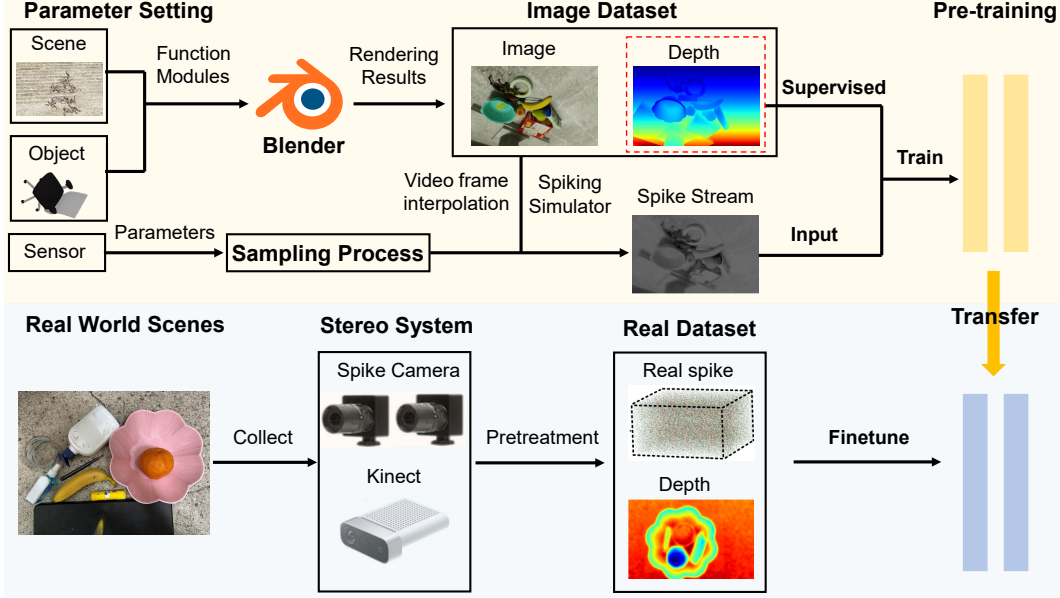


Figure 1: The overall pipeline is illustrated in the figure below. The upper path represents the training and evaluation process on synthetic dataset, while the lower path shows the transfer learning and testing procedure on real spike stream data by using the pre-trained model.

the sensor’s intrinsic advantages. Although event-based stereo methods have been developed for DVS cameras (e.g. [44–48]), they rely on temporal contrast, whereas spike cameras emit integrated intensity streams that require tailored processing techniques. Critically, the field lacks specialized algorithms and benchmarks designed for stereo depth estimation directly from raw spike streams, hindering progress in this promising field.

To bridge this gap, we introduce SpikeStereoNet (Fig. 1), an end-to-end, biologically inspired framework for generating stereo depth directly from raw spike streams. SpikeStereoNet integrates a recurrent spiking neural network (RSNN) into an iterative refinement loop [49] and models neuronal interactions [50] to capture the spatiotemporal dynamics inherent in spike data. The analysis of neuronal dynamics confirms the temporal stability, convergence, and expressive feature separation of the RSNN components. We introduce two novel benchmark datasets: a large-scale synthetic dataset with diverse scenes and ground-truth depth, and a real-world dataset containing synchronized stereo spike streams and corresponding depth sensor measurements. Experimental results demonstrate that SpikeStereoNet generalizes effectively, maintaining robust performance even when trained with substantially limited data. Our main contributions are as follows:

- We present the first large-scale synthetic and real-world raw spike stream datasets for stereo depth estimation to offer an evaluation benchmark for this emerging field.
- We propose a novel biologically inspired RSNN-based SpikeStereoNet architecture that refines asynchronous spike data through iterative updates.
- We analyze the dynamics of RSNN iterations to demonstrate the stability and convergence properties of the model.
- We demonstrate the data efficiency of the proposed framework, highlighting the strong generalization even with limited training samples.

2 Related Work

Spike Cameras and its Applications. Neuromorphic cameras, including event cameras [51–53] and spike cameras [54, 55], are inspired by the primate retina and operate asynchronously at the pixel level, providing ultra-high temporal resolution, wide dynamic range, low latency, and reduced energy consumption. Unlike event cameras, which follow a differential sampling model and capture

only luminance changes in logarithmic space, spike cameras employ an integral sampling model, accumulating photons until a threshold is reached and then firing spikes. Although event cameras tend to generate sparser output, they might lose scene textures, especially in static regions due to their change-based design. In contrast, spike cameras can capture richer spatial details, making them well suited for tasks such as high-speed imaging [23, 26, 56, 57], 3D reconstruction [58], motion deblurring [43, 59], optical flow [24, 25], object detection [60], occlusion removal [61] and monocular depth estimation [21].

Frame-Based Stereo Depth Estimation. Deep learning has significantly advanced stereo matching, leading to various methods with enhanced accuracy and generalization. The early methods [62–65] constructed cost volumes for each disparity and applied 3D CNNs for refinement. In contrast, recurrent refinement-based models [31–34, 36] inspired by RAFT-Stereo [29, 49] iteratively update the disparity without building a full 4D volume. It restricts 2D flow to the 1D disparity dimension and uses multi-level convolutional Gated Recurrent Unit (ConvGRU) for broader receptive fields, demonstrating strong generalization. Successor models such as CREStereo [30], IGEV-Stereo [33] and Selective-Stereo [36] improve in-domain performance, but RAFT-Stereo retains advantages in zero-shot settings [29]. Recent hybrid methods [66] combine cost filtering and iterative refinement for better trade-offs. Meanwhile, transformer-based solutions [37] utilize cross-attention or global self-attention. As deep stereo matching evolves, fully leveraging vision foundation models remains an open opportunity for real-world applications.

Event-Based Stereo Depth Estimation. Event cameras capture pixel-level brightness changes asynchronously, offering significant advantages over conventional frame-based cameras. Recently, event-based stereo depth estimation has advanced rapidly, with notable methods exploiting camera velocity [45] or deriving depth without explicit event matching [67]. Deep learning approaches propose novel sequence embedding [68] and fuse frame and event data [69, 70] to improve depth estimates in challenging scenarios. Moreover, some works [71] leverage standard models from the image domain, exploiting the inherent connection between frame and event data to improve stereo depth estimation performance.

3 Methods

This section describes the overall architecture of our proposed model (Fig. 2). Figs. 2a, b illustrate the biologically inspired framework design, where stereo signals from both cameras are processed through a context network analogous to the visual pathway. Correlation features are computed and temporally refined via recurrent spiking neural network (RSNN) blocks, mimicking depth perception mechanisms in the brain. Fig. 2c shows the overall network architecture and consists of four main components: a multi-scale spike feature extractor, a correlation volume construction module, a biologically inspired RSNN-based update operator, and a final upsampling module for high resolution disparity refinement.

Spike Camera Model. The spike camera, mimicking the retinal fovea, consists of a $H \times W$ pixel array in which each pixel asynchronously fires spikes to report luminance intensity. Specifically, each pixel independently integrates the incoming light over time. At a given moment t , if the accumulated brightness at pixel (i, j) reaches or exceeds a fixed threshold C , a spike is fired and the accumulator is reset: $A(i, j, t) = \int_{t_s}^t I(i, j, \tau) d\tau \geq C$, where $i, j \in \mathbb{Z}, i \leq H, j \leq W$. Here, $A(i, j, t)$ denotes the accumulated brightness at time t , $I(i, j, \tau)$ is the instantaneous brightness at the pixel (i, j) , and t_s is the last firing time of the spike. If t corresponds to the first spike, then $t_s = 0$. Due to circuit-level constraints, spike readout times are quantized. Although firing is asynchronous, all pixels periodically check their spike flags at discrete times $n\Delta t$ ($n \in \mathbb{Z}$), where Δt is a short interval on the order of microseconds. As a result, each readout forms a binary spiking frame of size $H \times W$. Over time, these frames constitute a $H \times W \times N$ binary spike stream:

$$S(i, j, n\Delta t) = \begin{cases} 1 & \text{if } \exists t \in [(n-1)\Delta t, n\Delta t] \text{ s.t. } A(i, j, t) \geq C, \\ 0 & \text{if } \forall t \in [(n-1)\Delta t, n\Delta t] \text{ s.t. } A(i, j, t) < C, \end{cases} \quad (1)$$

then we denote the generated spike stream as $S \in \{0, 1\}^{N \times H \times W}$, where H and W signify the height and width of the image, and N represents the temporal steps of the spike stream.

Spike Feature Extraction. The structure of spike feature extraction is illustrated in Fig. 3(a), which outlines the architecture of the feature network or the context network. The feature network

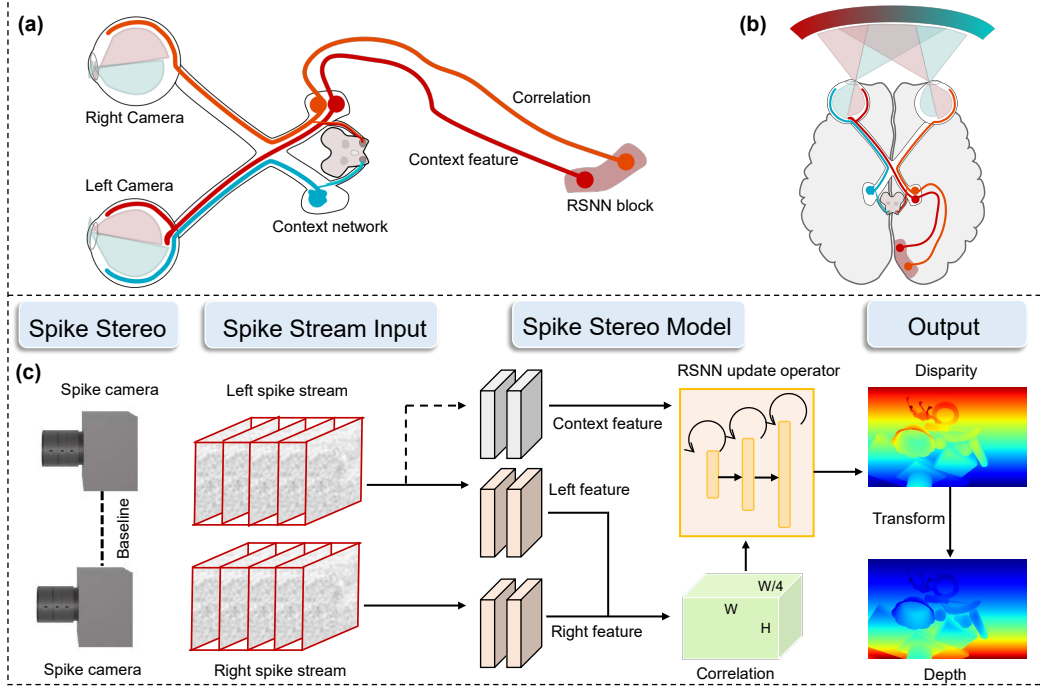


Figure 2: The pipeline of the proposed solution. (a) and (b) The components of the biological visual system, each corresponding to a specific module in the computational framework. (c) The overview of SpikeStereoNet: Multi-scale spike features are first extracted to construct a correlation pyramid, followed by a biologically inspired RSNN-based update operator that iteratively refines disparity using local cost volumes and contextual cues. The final disparity map are upsampled to produce high-resolution depth estimates.

receives the left and right input spike streams $S_{l(r)} \in \mathbb{R}^{N \times H \times W}$, applying a 7×7 convolution to downsample them to half-resolution. Then, a series of residual blocks is used for spike feature extraction, followed by another downsampling layer to obtain 1/4-resolution features. For more expressive representations, we further generate multi-scale features at 1/4, 1/8, and 1/16 scales, denoted $\{f_{l,i}, f_{r,i} \in \mathbb{R}^{C_i \times H_i \times W_i}\}, i \in \{4, 8, 16\}$. Among these, $f_{l,4}$ and $f_{r,4}$ are used to construct the cost volume, while all levels are used as a guide for the 3D regularization network.

The context network shares a similar backbone with the feature network. It consists of residual blocks and additional downsampling layers to produce multi-scale context features $f_{c,i}$ at 1/4, 1/8, and 1/16 of the original resolution, each with 128 channels. These features are used to initialize and inject context into the RSNNs at each recurrent iteration.

Correlation Volume. Given the left and right feature maps $f_{(l)}$ and $f_{(r)}$, we first construct a correlation cost volume of all pairs: $C_{ijk} = \sum_h f_{(l)hij} \cdot f_{(r)hik}$. Next, we build a 4-level correlation pyramid $\{C_i\}_{i=1}^4$ applying 1D average pooling along the last dimension. Each level is obtained using a kernel size of 2 and a stride of 2, progressively downsampling the disparity dimension.

RSNN-Based Update Operator. To capture multi-scale spatial and temporal information, a three-layer RSNN is used as the update module. Each RSNN layer contains recurrent intra-layer connections and feedforward inter-layer connections to facilitate hierarchical temporal processing (Fig. 3b, c). At 1/8 and 1/16 resolutions, RSNN modules receive inputs from the context features, the postsynaptic currents obtained by weights from the spike states at the same layer and the previous layer. These recurrent and feedforward connections expand the receptive field and facilitate feature propagation across scales. At 1/4 resolution level, the model additionally receives the current disparity estimate and the local cost volume derived from the correlation pyramid. A single RSNN updates as follows (the default parameters are the l -th layer network, the parameters from the previous layer are explicitly

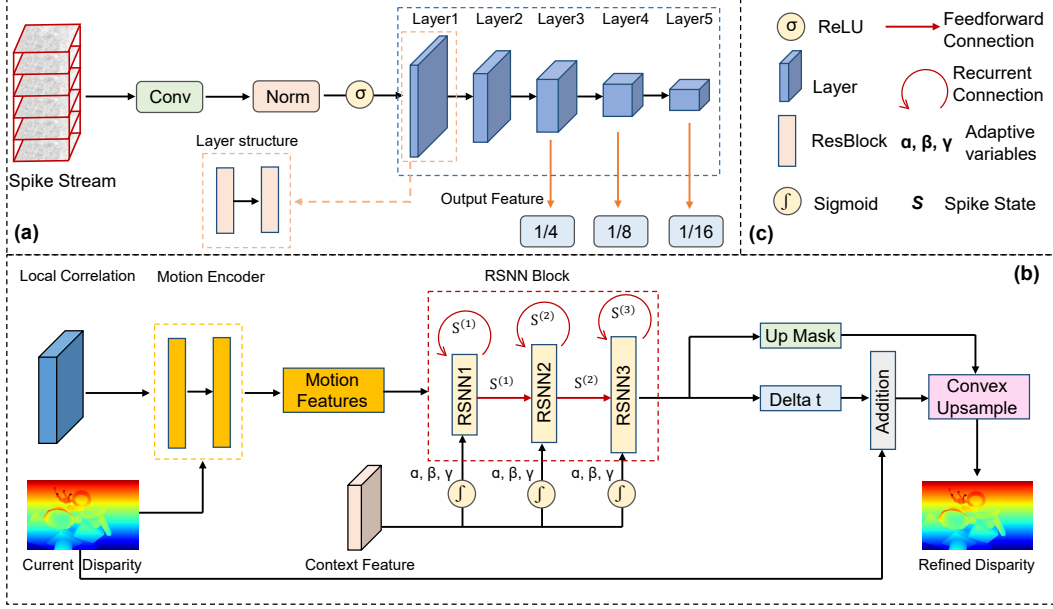


Figure 3: Illustration of the detailed structure of spike feature extraction and the RSNN-based update module. **(a)** Spike feature extraction: It comprises one context network and two feature networks, which extract multi-scale correlation features, contextual features, and the initial hidden state from the spike streams. A single network structure is illustrated in the diagram. **(b)** RSNN-based update block: Local correlations and disparity fields are used to generate motion features, which update the RSNN hidden states through recurrent and feedforward connections. The RSNN at the highest resolution is responsible for refining the disparity estimates. **(c)** Descriptions of key modules.

indicated, and bold symbols represent vectors/matrices):

$$\begin{aligned}
 \alpha_t &= \sigma(\text{Conv}([s_{t-1}, x_t], W_\alpha) + c_\alpha), \\
 \beta_t &= \sigma(\text{Conv}([s_{t-1}, x_t], W_\beta) + c_\beta), \\
 \gamma_t &= \sigma(\text{Conv}([s_{t-1}, x_t], W_\gamma) + c_\gamma), \\
 s_t &= f(s_{t-1}, s_t^{(l-1)}, \alpha_t, \beta_t, \gamma_t),
 \end{aligned} \tag{2}$$

where s_t is the spike state in layer l and timestep t , c_α , c_β , and c_γ are context embeddings from the context network, and σ is the sigmoid function. The number of channels in the hidden states of RSNNs is 128, same as in the context feature. The function $f(\cdot)$ corresponds to our proposed adaptive leaky integrate-and-fire neuron model, defined as:

$$\begin{aligned}
 h_t &= \alpha_t \cdot v_{t-1} + (1 - \alpha_t) \cdot (W_{\text{rec}} s_{t-1} + W_f s_t^{(l-1)}), \\
 v_t^{\text{th}} &= \beta_t \cdot v_{\text{peak}}, \\
 s_t &= \theta(h_t - v_t^{\text{th}}), \\
 v_t &= h_t - \gamma_t \cdot s_t \cdot v_t^{\text{th}},
 \end{aligned} \tag{3}$$

where v_t is the membrane potential, v_t^{th} is the firing threshold and $\theta(\cdot)$ is the Heaviside step function. W_{rec} and W_f are the recurrent and feedforward synaptic weights, respectively, implemented as convolutional kernels. And α , β and $\gamma \in [0, 1]$ are adaptive variables, representing retention of membrane potential, firing threshold, and soft reset in the previous step, respectively. This update mechanism draws biological inspiration from neuroscience findings showing that key neuronal properties such as firing threshold, resting potential, and membrane time constant are not fixed but vary dynamically in biological neural circuits depending on neuronal state and contextual input.

Spatial Upsampling. Based on the hidden state of the final layer, the network predicts both residual disparity through two convolutional layers and then updates the current disparity: $d_t = d_{t-1} + \Delta d_t$. Finally, the disparity map at 1/4 resolution is upsampled to the full resolution using a convex combination strategy.

Loss Function. We supervise the network for a composite loss function composed of three terms: the main loss, the firing rate regularization, and the voltage regularization. The overall loss is defined as:

$$\begin{aligned} L &= L_{\text{stereo}} + \lambda_f L_{\text{rate_reg}} + \lambda_v L_{\text{v_reg}} \\ &= \sum_{t=1}^T \eta^{T-t} \|d_{\text{gt}} - d_t\|_1 + \lambda_f \sum_{i=1}^N (r_i - r_0)^2 + \lambda_v \sum_{i=1}^N \sum_{t=1}^T v_i(t)^2, \quad \eta = 0.9. \end{aligned} \quad (4)$$

The first term L_{stereo} is the main loss, which is the L_1 - norm distance between all predicted disparities $\{d_t\}_{t=1}^T$ and the ground truth disparity d_{gt} with increasing weights. To constrain the firing rate of neurons and promote temporal sparsity, the second term $L_{\text{rate_reg}}$ is firing rate regularization, r_i is the average firing rate of the i -th neuron during time T , and r_0 is the target firing rate. Furthermore, the third term $L_{\text{v_reg}}$ serves as the voltage regularization term, where N represents the total number of neurons. Both regularization terms encourage sparsity and benefit the performance.

4 Experiments

4.1 Datasets

Synthetic Dataset. To support supervised training for spike-based depth estimation, we construct a high-quality synthetic dataset using Blender [72]. The dataset includes 150 training scenes and 40 testing scenes, each containing 256 pairs of RGB images and the corresponding ground-truth depth maps. The dataset construction process [73, 74] and additional details are provided in the Supplementary Materials.

To generate dense spike streams with high temporal resolution in dynamic environments, we apply a video interpolation method [75] to synthesize 50 intermediate RGB frames between each pair of adjacent frames. These high-frequency image sequences are then converted into spike streams using a biologically inspired spike generation mechanism that models the analog behavior of real spiking camera circuits. Noise modeling is also integrated to further enhance realism. Our dataset provides synchronized high-resolution spike streams and pixel-wise depth ground truth, offering a comprehensive and reliable benchmark for training and testing.

Real Dataset. We collect a real-world spike stereo dataset using two spike cameras and one depth camera (Kinect) in diverse environments to evaluate the generalization and practical performance of the model. The spike cameras have a resolution of 400×250 with a temporal resolution of 20,000 Hz, while the depth camera provides RGB-D data at 1280×720 resolution and 30 FPS. The captured scenes involve various objects of the real world recorded in indoor settings. The dataset includes over 2000 stereo spike stream pairs (more than 200,000 frames) and depth maps, and is divided into training and test sets. Further details about our real dataset and the spike stereo binocular system are provided in the Supplementary Materials.

4.2 Implementation Details

We implement SpikeStereoNet using PyTorch and perform all experiments on NVIDIA RTX 4090 GPUs. The model is trained using the AdamW optimizer with gradient clipping in the range of $[-1, 1]$. A one-cycle learning rate schedule is adopted, with an initial learning rate set to 2×10^{-4} . During training, we apply 16 update iterations for each sample. We randomly cropped the spike stream to 224×384 spatially during the training procedure and set the batch size as 8 for 300k steps. We use random horizontal and vertical flips as data augmentation to balance the motion in the training dataset.

4.3 Experimental Results

Our method is compared with various categories of stereo depth estimation approaches. First, event-based stereo networks are compared, such as ZEST [47]. Next, state-of-the-art frame-based stereo networks are evaluated, including Selective-Stereo [36], RAFT-Stereo [29], and GMStereo [37], etc. These methods cover different strategies, such as iterative refinement or transformer-based global matching. All networks are retrained using the same settings and evaluated under identical conditions to ensure fair comparison.

Table 1: Quantitative results on the testing set of synthetic dataset, and all methods use spike streams as input. Best results for each evaluation metric are bolded, second best are underlined. Lower is better for all metrics.

Method	bad 1.0 ↓ (%)	bad 2.0 ↓ (%)	bad 3.0 ↓ (%)	AvgErr ↓ (px)	Params (M)	FLOPs (B)
CREStereo [30]	16.12	10.54	4.40	0.81	5.43	863
RAFT-Stereo [29]	9.67	4.64	2.76	0.48	11.41	798
GMStereo [37]	17.61	11.05	4.39	0.83	<u>7.40</u>	160
IGEV-Stereo [33]	12.50	6.50	4.27	0.76	12.77	614
DLNR [31]	10.15	4.67	2.81	0.55	57.83	1580
ZEST [47]	11.10	4.94	3.50	0.62	340.75	989
MoCha-Stereo [39]	11.93	5.67	4.39	0.73	20.96	935
Selective-Stereo [36]	9.24	<u>4.57</u>	2.66	<u>0.45</u>	13.32	957
MonSter [40]	<u>9.23</u>	4.64	2.72	0.46	388.69	1567
SpikeStereoNet (Ours)	9.22	4.52	<u>2.68</u>	0.44	12.15	<u>537</u>

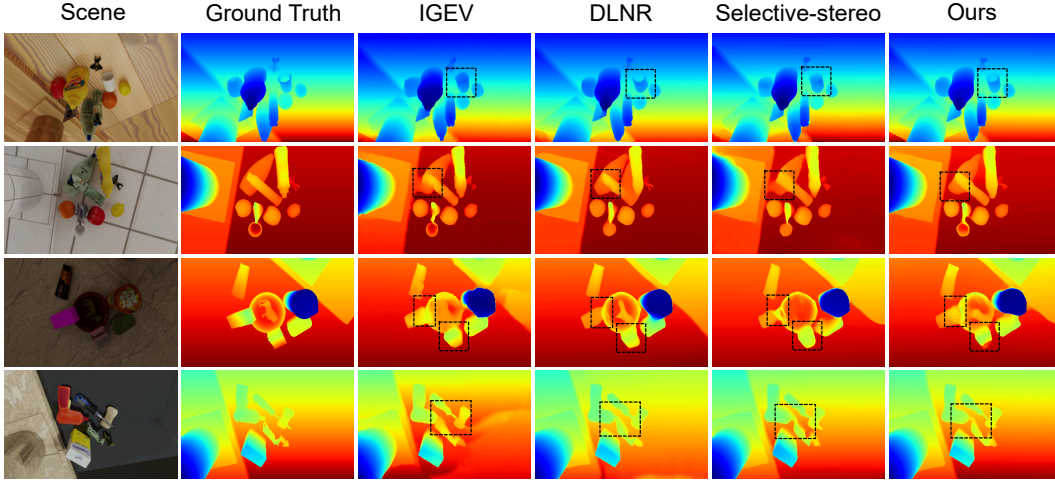


Figure 4: From left to right: synthetic scene images from the left view, ground-truth depth, depth prediction results from existing stereo methods and our method.

Experiments on the Synthetic Dataset. We use the average end-point error (AvgErr) and the bad pixel ratio with different pixel thresholds to evaluate the quality of stereo depth estimation. The results of the quantitative comparisons are presented in Table 1. As shown in the table, we achieve the best performance among the listed methods for almost all metrics in the synthetic dataset. Fig. 4 illustrates the scene, the spike stream, the ground truth of the depth, and the predicted depth maps of different methods. Our method significantly outperforms competing approaches, producing more accurate and sharper depth results. The sharpness and density of the predicted depth maps demonstrate the effectiveness of our approach in the spike stream input, especially in complex and cluttered scenes.

In addition to accuracy metrics, Table 1 also reports the params and floating-point operations (FLOPs) for each method. SpikeStereoNet achieves a favorable balance between performance and efficiency, ranking second in FLOPs while outperforming most models in parameter number. This demonstrates that the additional computational cost, incurred to better capture the spatiotemporal characteristics and asynchrony of spike data is justified. Compared to other methods that handle complex temporal structures, such as Transformer-based approaches, SpikeStereoNet remains highly competitive in terms of efficiency. Additional comparisons and analyses are provided in Supplementary Materials.

Experiments on the Real Dataset. To bridge the domain gap between synthetic and real data, we apply a domain adaptation strategy to fine-tune our model, which is initially trained on synthetic datasets. The visualization results are shown in Fig. 5. Compared with other methods, our approach produces more accurate depth maps with sharper object boundaries and finer structural details. It

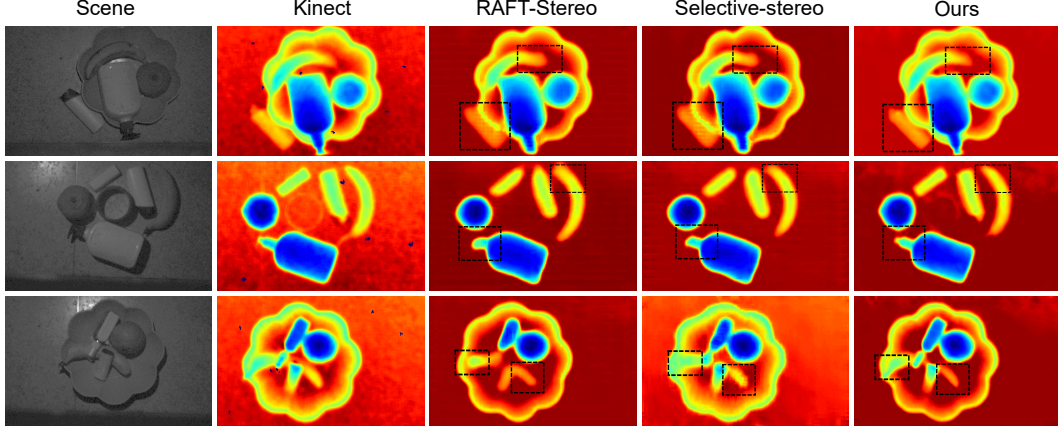


Figure 5: Visual results of our method and competing approaches on the real dataset. The "Scene" refers to the gamma-transformed temporal average of spike streams. "Kinect" represents the raw depths captured by the depth camera.

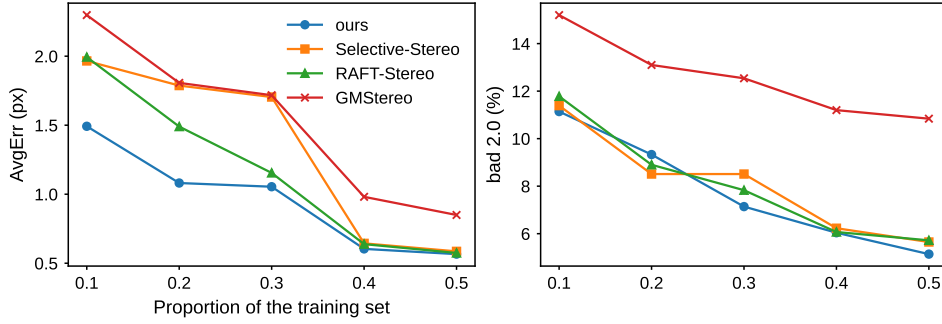


Figure 6: Data efficiency experiments on the synthetic dataset. The horizontal axis indicates the proportion of the training set, while the vertical axes represent the AvgErr and 2 px error, respectively.

performs especially well in challenging regions, such as textureless surfaces and high-illumination areas, benefiting from the integration of spike domain characteristics. These results highlight the effectiveness of our domain adaptation strategy and the robustness of our model in real-world stereo depth estimation from spike streams. More quantitative comparison experiments and analyses are accessible in the Supplementary Materials.

Data Efficiency. To evaluate the generalizability of our model, we conduct experiments by training it on randomly selected subsets of the synthetic dataset, using 10% to 50% of the full training data in 10% increments. All models, including ours and the baselines, are tested on the complete testing set to ensure a fair and consistent comparison. As illustrated in Fig. 6, our model consistently outperforms other methods across all training data ratios. The performance gap becomes more significant as the training data size decreases, demonstrating that our model generalizes better in data-scarce scenarios. These results highlight the robustness and efficiency of our approach in learning meaningful representations, even with limited supervision.

Explanation of Iterative Refinement Structures. We analyze the dynamic behavior of our hybrid neural network to assess its temporal characteristics. As shown in Fig. 7, the left panel illustrates that hidden state differences decrease over time, indicating convergence. The middle panel displays the eigenvalue spectrum of the Jacobian matrix at the final time step, with all eigenvalues lying within the unit circle, which confirms the stability of the system. The right panel presents a PCA of hidden states across input batches, showing increasing dispersion over time, which reflects the model's ability to encode diverse temporal patterns. These results highlight the temporal stability, robustness, and expressive power of the network.

Beyond validating our specific RSNN design, this dynamics analysis sheds light on why iterative refinement, the core mechanism in architectures like RAFT-Stereo [29], is so effective. Our results

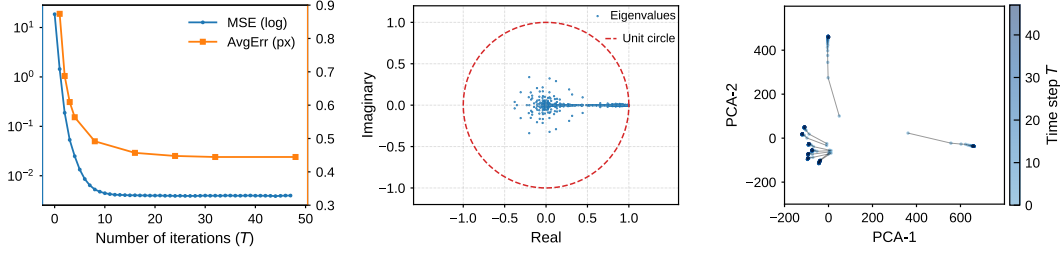


Figure 7: Visualization of network dynamics analysis. The sub-figures show, in order: the change in hidden state differences over time in the RSNN, the eigenvalue spectrum of the network for stability analysis, and the PCA of different inputs over time.

Table 2: Ablation studies of our proposed modules. All experiments are conducted on the synthetic dataset.

Setting of experiment	bad 2.0 (%) ↓	bad 3.0 (%) ↓	AvgErr (px) ↓
Removing RC and FFC module	12.07	6.46	1.29
Removing RC module	11.05	4.48	0.83
Removing FFC module	5.86	3.63	0.68
Full model	4.52	2.61	0.44
Removing Reg module	6.38	2.85	0.51
Removing GN module	7.33	2.97	0.57
Removing Reg and GN module	7.77	3.41	0.61

explicitly show how a well-behaved iterative updater can maintain stability while progressively integrating information over time steps to refine the estimate towards the correct solution. This provides a dynamical grounding for the empirical success of iterative methods.

4.4 Ablation Study

To evaluate the effectiveness of each component in our framework and analyze their contributions to the overall performance, we conduct a series of ablation studies in the synthetic dataset.

Connection Structure. We analyze the effect of connection structures within RSNN. When using feedforward connections (FFC) between layers or recurrent connections (RC) within layers, there is a slight increase in runtime and parameter count. However, this structural enhancement leads to significantly better overall performance. As shown in the upper part of Table 2, incorporating these connections results in improved accuracy in all evaluation metrics. The improved information flow across time and layers enables networks to capture richer temporal dynamics and finer spatial details, highlighting the importance of these connections in achieving accurate stereo depth estimation.

Regularization. We investigate the impact of removing regularization (Reg) terms and group normalization (GN). As listed in the lower part of Table 2, removing either the regularization on firing rate and membrane potential or the group normalization applied to adaptive input variables leads to a clear drop in performance. The absence of regularization results in unstable training and increased prediction error, while removing group normalization slows convergence and reduces final accuracy. When both components are removed simultaneously, performance is further degraded.

5 Conclusion

We present SpikeStereoNet, the state-of-the-art deep learning framework for stereo depth estimation from spike streams of spike camera, with a biologically plausible update operator based on recurrent spiking neural networks (RSNNs). In addition, we construct synthetic and real-world stereo spike stream datasets with corresponding depth maps. Extensive experiments demonstrate that our method achieves SOTA performance, benefiting from the data efficiency and convergence of RSNN. Overall,

SpikeStereoNet represents an advance in neuromorphic stereo vision and demonstrates the potential of spike-based systems for efficient, accurate, and high-speed 3D perception.

Limitations. Our method is limited by the small scale of the real world dataset and the domain gap between the synthetic and real data. Although domain adaptation helps, its effectiveness is constrained by the quality of real labeled data and cannot fully bridge distribution differences. We plan to extend our method to handle these questions in future work.

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A Technical Appendices and Supplementary Material

The supplementary materials provide additional implementation details and extended experimental results for the SpikeStereoNet framework introduced in the main paper. It is organized into four distinct sections: Theory Analysis and Method in Appendix A.1, Details of the Synthetic Dataset in Appendix A.2, Details of the Real Dataset in Appendix A.3, More Experimental Results in Appendix A.4 and Broader Impacts in Appendix A.5.

A.1 Theory Analysis and Method

A.1.1 Problem Statement

Given a continuous binary spike stream $S_N^{t_1} \in \{0, 1\}^{H \times W \times N}$ with a spatio-temporal resolution of $H \times W \times N$, centered on the timestamp t_1 , we divide it into three non-overlapping sub-streams $S_N^{t_0}$, $S_N^{t_1}$, and $S_N^{t_2}$, centered on timestamps t_0 , t_1 , and t_2 , respectively. The goal is to estimate the disparity map corresponding to t_1 . Here, $S_N^{t_1}$ provides the essential spatial structure for estimating the disparity, while $S_N^{t_0}$ and $S_N^{t_2}$ provide temporal cues to improve consistency and robustness. Letting $S_n \in \{0, 1\}^{W \times H}$ denote the n -th spike frame, we predict the disparity at timestamp t_1 from the full spike stream $\{S_n\}, n = 0, \dots, k$, leveraging both spatial features and temporal continuity.

A.1.2 Noise Simulation

Noise in spiking cameras originates mainly from the dark current. The accumulation of brightness at pixel (i, j) over time t can be expressed as:

$$A(i, j, t) = \int_{t_{i,j}^{\text{pre}}}^t (I(i, j, \tau) + I_{\text{dark}}(i, j, \tau)) d\tau, \quad (5)$$

where $A(i, j, t)$ represents the integrated brightness, $t_{i,j}^{\text{pre}}$ is the previous firing time for pixel (i, j) , and I_{dark} denotes the random noise component caused by dark current. In our synthetic dataset, the dark currents in the circuits introduce thermal noise, which is the type of noise modeled in this work.

A.1.3 Video-to-Spike Preprocessing Pipeline

We introduce a two-stage preprocessing pipeline to convert conventional image data into temporally precise spike streams, comprising neural network-based frame interpolation and spike encoding.

Frame Interpolation for Enhanced Temporal Resolution. To enhance temporal resolution, raw frames from dynamic synthetic datasets are processed using a pre-trained video frame interpolation model. The architecture comprises a hierarchical feature extractor, a MultiScaleFlow module for multi-resolution optical flow estimation, and a U-Net for residual refinement through bidirectional flow guidance and mask fusion.

For temporal expansion, we applied a frame rate upsampling factor of $\times 50$ to the synthetic dataset. The output is represented as a 4D tensor of shape $[T, H, W, C]$, where T is the temporal length, $H \times W$ is the spatial resolution, and $C = 3$ denotes RGB channels.

Spike Encoding via Temporal Integration. The high-frame-rate RGB sequences are converted to spike streams through a temporal integration algorithm:

1. Convert RGB frames to grayscale and normalize pixel intensities to $[0, 1]$.
2. Accumulate membrane potential over time as $V_t = V_{t-1} + I_t$.
3. Spike generation:

$$\text{spike matrix}[t, x, y] = \begin{cases} 1 & \text{if } V_t(x, y) \geq \theta \\ 0 & \text{otherwise} \end{cases}$$

where threshold $\theta = 5.0$ and potential reset $V_t \leftarrow V_t - \theta$ after spike.

4. Repeat until all frames processed.

Spike streams are stored as binary tensors of shape $[T, H, W]$ using the `StackToSpike` function with configurable noise I_{noise} and threshold θ . For compact storage and compatibility for our framework, the `SpikeToRaw` module compresses spikes (8 per byte) into .dat format, enabling lossless reconstruction during inference.

The proposed two-stage preprocessing pipeline effectively bridges conventional images and neuromorphic vision processing. By combining deep learning-based frame interpolation with bio-inspired spike encoding, we achieve:

- **Temporal Super-Resolution:** Neural interpolation extends temporal sampling density by 50× through multi-scale optical flow and attention mechanisms, preserving physical consistency in dynamic scenes.
- **Biologically Plausible Encoding:** The temporal integration algorithm emulates retinal neuron dynamics, converting intensity variations into sparse spike events with adaptive threshold control.
- **System Compatibility:** Serialized spike data (.dat) with byte-level compression and structured formatting ensure seamless integration into brain-inspired stereo depth estimation systems.

This pipeline facilitates the efficient transformation of dynamic scene datasets into spike-compatible formats, preserving configurable spatio-temporal characteristics and laying a solid foundation for spike-based stereo depth estimation.

To assess the fidelity of the dataset conversion, we utilize the Texture From Interval (TFI) algorithm of the SpikeCV [76] library to reconstruct grayscale images from spike streams with dimensions $[T, H, W]$. This method exploits the spatiotemporal sparsity and encoding capacity of spike streams to approximate the textural structure of conventional images.

The core idea behind TFI is that the temporal interval between adjacent spikes encodes local texture intensity: shorter intervals correspond to higher pixel activity and brighter intensity. The algorithm identifies the two nearest spike timestamps within a bounded temporal window ($\pm \Delta t$) for each pixel and computes the pixel-wise grayscale value based on their interval duration.

A.1.4 Adaptive Leaky Integrate-and-Fire Neuron Model

Common RSNN models often overlook several biologically relevant dynamic processes, particularly those occurring over extended time scales. To address this, we incorporate neuronal adaptation into our RSNN design. Empirical studies, such as those in the Allen Brain Atlas, indicate that a significant proportion of excitatory neurons exhibit adaptation with varying time constants. Several approaches to fit models for adapting neurons to empirical data.

In this work, we adopt a simplified formulation. The membrane potential of each neuron is updated using the Exponential Euler Algorithm [77]. The firing threshold v_t^{th} increases by a fixed value β_t following each spike, while the peak voltage $v_t^{th} = \beta_t \cdot v_{peak}$ remains constant. At a discrete time step of $\Delta t = 1$ ms, the neuronal membrane potential update rule is defined as:

$$h_t = \alpha_t \cdot v_{t-1} + (1 - \alpha_t) \cdot (W_{rec}s_{t-1} + W_f s_t^{(l-1)}), \quad (6)$$

where $\alpha_t = \exp(-\Delta t / \tau_a)$ and $s \in \{0, 1\}$ denotes the binary spike train of neuron. The retention of membrane potential is controlled by a decay factor γ_t . In summary, the neuron model incorporates three adaptive parameters that govern the decay of membrane potential, firing threshold, and reset potential, jointly determining the neuron’s adaptive dynamics, respectively.

A.1.5 Applying BPTT to RSNN

Although backpropagation through time (BPTT) is not biologically plausible, it serves as an effective optimization strategy for training RSNNs over extended temporal sequences, analogous to how evolutionary and developmental processes adapt biological systems to specific tasks. Prior work has applied backpropagation to feedforward spiking neural networks by introducing a pseudo-derivative to approximate the non-existent gradient at spike times. This surrogate derivative increases smoothly from 0 to 1 and then decays, allowing gradient-based learning. In our implementation, the pseudo-derivative’s amplitude is attenuated by a factor less than 1, which improves BPTT stability and performance when training RSNNs over longer time horizons. Throughout, we use the following surrogate function $g(x)$:

$$g(x) = \text{sigmoid}(\alpha x) = \frac{1}{1 + e^{-\alpha x}}, \quad (7)$$

the spiking function of sigmoid gradient is used in backpropagation. The gradient is defined by:

$$g'(x) = \alpha * (1 - \text{sigmoid}(\alpha x)) * \text{sigmoid}(\alpha x), \quad (8)$$

where α is a parameter that controls the slope of the surrogate derivative. Unless mentioned otherwise we set $\alpha = 4$.

A.1.6 Detail of Loss Function

In our training framework, the total loss L is composed of three distinct components, each serving a specific purpose in optimizing the performance of model. Specifically, the total loss is defined as

$$L = L_{\text{stereo}} + \lambda_f L_{\text{rate_reg}} + \lambda_v L_{v_reg}, \quad (9)$$

where L_{stereo} represents the stereo loss, $L_{\text{rate_reg}}$ denotes the firing rate regularization loss, and $L_{\text{v_reg}}$ corresponds to the velocity regularization loss. The parameters λ_f and λ_v are regularization coefficients that control the influence of the respective regularization terms. The stereo loss L_{stereo} is designed to measure the discrepancy between the ground-truth disparity d_{gt} and the predicted disparity d_t over time t . It is formulated as follows:

$$L_{\text{stereo}} = \sum_{t=1}^T \eta^{T-t} \|d_{\text{gt}} - d_t\|_1, \quad (10)$$

where $\eta = 0.9$ is a discount factor that assigns more weight to recent disparities, ensuring that the model focuses more on recent predictions.

The firing rate regularization loss $L_{\text{rate_reg}}$ aims to stabilize the firing rates of neurons by penalizing deviations from the target firing rate r_0 . It is defined as:

$$L_{\text{rate_reg}} = \sum_{i=1}^N (r_i - r_0)^2, \quad r_i = \frac{1}{T} \sum_{t=1}^T S_i(t), \quad (11)$$

where r_i is the average firing rate of neuron i over time T , this term encourages the model to maintain consistent firing rates across neurons, thereby enhancing stability and preventing over-activation.

The voltage regularization loss $L_{\text{v_reg}}$ is introduced to regularize the temporal dynamics of the model by penalizing high velocities of neurons. It is expressed as:

$$L_{\text{v_reg}} = \sum_{i=1}^N \sum_{t=1}^T v_i(t)^2, \quad (12)$$

where $v_i(t)$ represents the membrane potential of neuron i at time t , and N is the number of neurons in the network. This regularization term helps smooth the temporal behavior of the model, ensuring that changes in neuronal states are gradual and controlled.

Overall, the combination of these loss components allows for a comprehensive optimization strategy that balances accuracy, firing rate stability, and temporal smoothness, thereby enhancing the robustness and performance of the model in stereo vision tasks.

A.2 Details of the Synthetic Dataset

A conventional active stereo depth system typically comprises an infrared (IR) projector, a pair of IR stereo cameras (left and right) and a color camera. Depth measurement is achieved by projecting a dense IR dot pattern onto the scene, which is subsequently captured by the stereo IR cameras from different viewpoints. Depth values are then estimated by applying a stereo matching algorithm to the captured image pair, leveraging the disparity between them. We use the method from [73, 74] replicates this process, containing light pattern projection, capture, and stereo matching. The simulator is implemented based on the Blender [72] rendering engine.

A.2.1 Random Scenes

For each scene, we generate the corresponding '.meta' files to store structured metadata information. Additionally, for each object instance, a high-quality '.obj' mesh file is created to represent its geometry. To simulate realistic camera motion, a continuous trajectory is generated by sampling camera poses uniformly from a virtual sphere with a fixed radius of 0.6 meters. The representative camera viewpoints are separated by a constant angular increase of 3° , ensuring smooth transitions along the trajectory. The objects are rotated accordingly to maintain consistent spatial alignment. Furthermore, a uniform background and consistent lighting configuration are applied across all scenes to reduce domain-specific biases and enhance rendering consistency. To further bridge the domain gap between synthetic and real-world data, we adopt domain randomization strategies by randomizing object textures, material properties (from specular, transparent, to diffuse), object layouts, floor textures, and lighting conditions aligned with camera poses.

A.2.2 Stereo Matching

We perform stereo matching to obtain the disparity map, which can be transferred to the depth map leveraging the intrinsic parameters of the depth sensor. Specifically, a matching cost volume is constructed along the epipolar lines between the left and right infrared (IR) images. The disparity at each pixel is determined by selecting the position with the minimum matching cost. To enhance accuracy, sub-pixel disparity refinement is performed via quadratic curve fitting. In addition, several post-processing techniques are employed to improve depth quality, including left-right consistency checks, uniqueness constraints, and median filtering. These steps collectively yield more realistic and robust depth estimations. The calculation method from disparity to depth is as follows:

$$Z = \frac{Bf}{d + (c_{x1} - c_{x0})}, \quad (13)$$

Table A1: Statistics of the training and test set of synthetic dataset.

	Scene Indexes	Number of Depth Map	Resolution	Number of Spike Frames	Number of Scenes
Training Dataset	0 – 149	38,250	400×250	76,500	150
Test Dataset	150 – 189	10,200	400×250	20,400	40

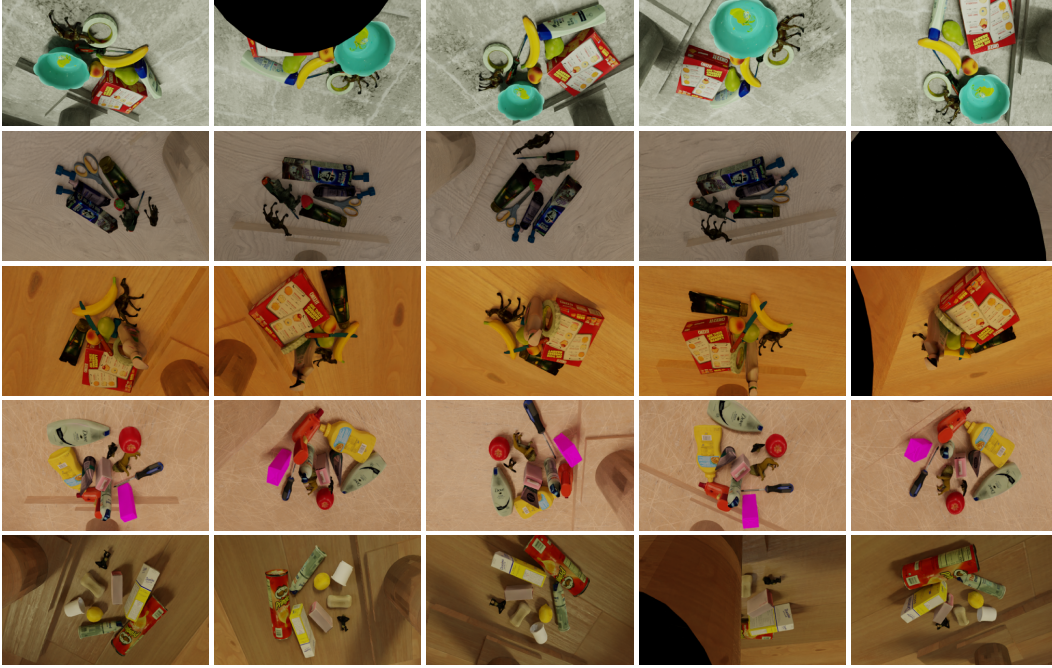


Figure A1: Scenes for generating the training set of the synthetic dataset. Selected five scenes, each displayed as a row in which the images correspond to different camera viewpoints of the same scene (partial views).

where Z is the obtained depth, d is the disparity, f denotes the camera focal length, B is the baseline distance between the stereo cameras, and $c_{x1} - c_{x0}$ represents the horizontal offset between the principal points of the right and left cameras.

A.2.3 Synthetic Dataset

We make use of domain randomization and depth simulation and construct the large-scale synthetic dataset. In total, the dataset consists of two subsets: (1) Training set: Consists of 76,500 spike streams and 38,250 depth maps generated from 150 scenes composed of 88 distinct objects, with randomized specular, transparent, and diffuse materials. (2) Test set: 20,400 spike streams and 10,200 depth maps generated from 40 new scenes containing 88 distinct objects. The scenes in the dataset are shown in Fig. A1 and Fig. A2, and the detailed statistics of dataset are shown in Table A1.

A.3 Details of the Real Dataset

We construct a stereo imaging system using two spike cameras and one RGB-D camera (Kinect) to simultaneously capture left/right spike streams and the corresponding ground-truth depth images. As illustrated in Fig. A3, the two spike cameras are mounted on a high-precision alignment rig to ensure accurate baseline calibration. The spatial resolution of the spike camera is 400×250 , and the temporal resolution is 20,000 Hz. The resolution of the RGB-D camera is 1280×720 , and the frequency is 30 fps. Compared with LiDAR, the RGB-D sensor provides dense depth annotations within its effective range, avoiding the sparse and incomplete depth output typically associated. To ensure spatial alignment between modalities, intrinsic and extrinsic calibrations are performed between the spike cameras and the depth camera. The captured depth maps are then geometrically warped to the coordinate frame of the left spike camera for accurate correspondence.



Figure A2: Scenes for generating the test set of the synthetic dataset. Selected two scenes, each displayed as a row in which the images correspond to different camera viewpoints of the same scene (partial views).



Figure A3: The photograph of our hybrid imaging system. The RGB-D camera is fixed on the top plane of the stereo spike cameras. The extrinsic parameters between the cameras are calibrated.

The real dataset comprises some video sequences captured under diverse environmental conditions, including varying ambient illumination levels (e.g., low, medium, and high brightness) and a range of dynamic scenarios characterized by distinct object arrangements. These scenes encompass different spatial configurations and relative motions of multiple objects (e.g., camera shake and object motion), introducing varied degrees of motion blur. Each blurry input frame is paired with 100 corresponding spike frames, forming a substantial collection of blur-spike pairs suitable for training and evaluation.

A.4 More Experimental Results

A.4.1 Additional Comparative Results on the Synthetic Dataset

We present a complete stereo depth estimation sequence for a single scene within the synthetic dataset, capturing the full motion trajectory of objects throughout the dynamic environment. As illustrated in Fig. A4, the predicted depth maps effectively reconstruct the continuous evolution of the scene, accurately reflecting both spatial structure and temporal consistency. These results demonstrate the model’s ability to preserve coherent geometry across time, validating its effectiveness in handling dynamic visual inputs. The supplementary materials in the zip file include continuously changing videos.

In addition, we conducted a qualitative comparison between SpikeStereoNet and several state-of-the-art methods on the synthetic dataset. Our method consistently produces more accurate and visually coherent depth maps in various scenes. Further qualitative results are provided to demonstrate the robustness and effectiveness of our approach.

A.4.2 Comparison on the Real Dataset

In the main experimental section, we have presented the qualitative results of several models on the real-world dataset. Here, we further report quantitative evaluation metrics to complement visual analysis. Specifically, we adopt the average end-point error (AvgErr) and the bad pixel ratio as our evaluation criteria. The detailed results of different models are summarized in Table A2.

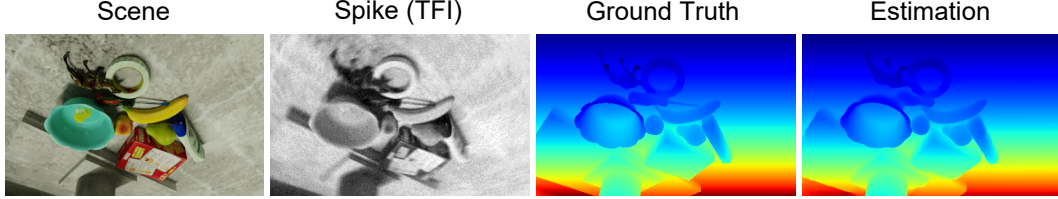


Figure A4: Video results of our method on the synthetic dataset in the single scene. From left to right, the figure shows the sequential scene images, the visualized spike stream using the TFI method, the ground-truth depth maps, and the depth estimation results produced by our proposed method.

Table A2: Quantitative results on the real dataset, and all methods use real spike streams as input.

Method	bad 2.0 (%) ↓	bad 3.0 (%) ↓	AvgErr ↓
RAFT-Stereo [29]	6.18	3.39	0.64
DLNR [31]	5.64	<u>3.38</u>	0.61
Selective-stereo [36]	<u>5.50</u>	3.43	<u>0.58</u>
SpikeStereoNet (ours)	5.33	3.19	0.56

As shown in the last row of Table A2, we can see that our method also outperforms other methods with the smallest AvgErr on the real dataset. The results of the experiment on the real dataset demonstrate that our method can effectively handle real scenarios.

A.4.3 Additional Results for Ablation Study

We provide additional ablation results to demonstrate the effectiveness of our designed modules. We analyze the impact of iteration steps by varying the number of iterations T in our RSNN-based model. As shown in Table A3, our method achieves competitive or superior performance with significantly fewer iterations compared to existing approaches. Unlike conventional GRUs or transformer-based update modules, RSNNs require fewer parameters and operations per iteration, thus maintaining a lower computational cost.

In addition, to further support the ablation studies presented in the main experimental section, we provide the corresponding qualitative results in Fig. A5. These visualizations clearly demonstrate the effectiveness of each module in our proposed framework and validate their individual contributions to the overall performance.

A.5 Broader Impacts

The proposed framework for stereo depth estimation from spike streams holds strong potential to advance neuromorphic vision in diverse application scenarios. In autonomous driving, the enhanced depth estimation enabled by spike-based sensing can improve 3D scene understanding, obstacle detection, and object localization, thereby contributing to more robust and reliable navigation systems. In the field of robotics, the fine-grained depth perception afforded by this method supports precise manipulation, mapping, and motion planning, especially in highly dynamic or high-speed environments where conventional cameras typically underperform. Moreover, this framework is the first that can directly estimate depth from raw stereo spike streams under supervised settings, enabling broader exploration of spike-based vision in both research and industrial applications.

Table A3: Ablation study for number of iterations and runtime.

	Number of Iterations (T)					
	2	3	4	8	16	32
AvgErr (px)	0.69	0.61	0.56	0.49	0.46	0.44
Runtime (s)	0.41	0.42	0.43	0.46	0.51	0.63

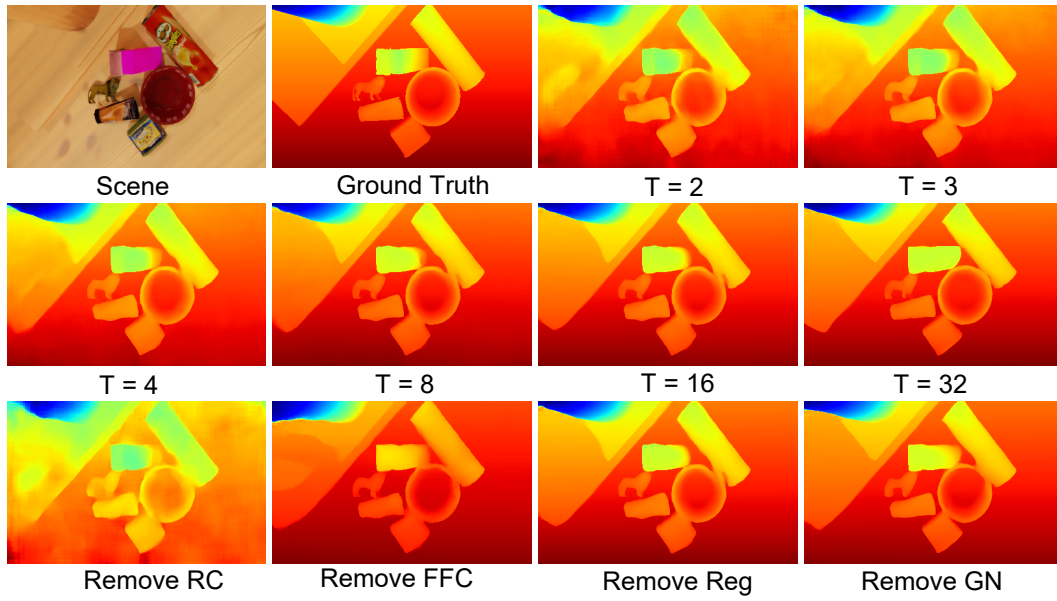


Figure A5: Visualization results of ablation study for evaluating the effectiveness of our framework on the synthetic dataset. Among them, $T = 32$ is the baseline of our model.